



# Huy Van Tran

AI ENGINEER | FULLSTACK ENGINEER

## Profile

Full-Stack Software Engineer with 3+ years building production web applications and AI-powered systems. Strong experience developing responsive React/TypeScript frontends and scalable Python/FastAPI backend services and REST APIs, backed by PostgreSQL and MongoDB. Hands-on with LLM-based applications and AI frameworks (LangGraph, RAG, function calling), computer vision, and data processing pipelines. Experienced across the full lifecycle — architecture, code review, containerization (Docker), and deployment — and comfortable working in international, collaborative environments.

**Core Stack:** React, TypeScript, Python, FastAPI, PostgreSQL, MongoDB, REST API, LangGraph/LangChain, RAG, LLM Integration, Docker, CI/CD (GitHub Actions), Git, Microservices

## Employment History

### Badminton Club Management Platform — AI Agent Layer, Personal Project

JAN 2026 — PRESENT

- Built a production AI agent (FastAPI + LangGraph) layered onto a React/Redux + Node.js app, letting admins manage club operations through natural language via LLM function calling over typed tools that wrap the existing Node.js REST routes.
- Secured write actions with a human-in-the-loop confirmation flow; agent carries the admin's JWT so it operates strictly within that admin's permissions.
- Added persistent conversation memory using LangGraph's MongoDB checkpointer, keyed per-admin and surviving restarts; built a multilingual (i18n) React frontend and bilingual (VI/EN) agent responses.
- Integrated an LLM-based pipeline (Grok / xAI) parsing free-form Vietnamese payment messages into structured fee data; deployed on DigitalOcean with Nginx + Gunicorn

**Tech Stack:** React | Redux | i18n | TypeScript | Node.js | Python | FastAPI | LangGraph | LLM Function Calling | Grok (xAI) | MongoDB | Docker | Nginx

### AI Patient Safety & Smart Hospital Platform, Enigco

JAN 2025 — PRESENT

- Lead the development of AI-powered healthcare solutions in a 3-member startup team.
- Built end-to-end AI features from data preprocessing, model training, backend APIs, to frontend integration.
- Developed computer vision models for seizure, fall detection, and abnormal patient behavior recognition.
- Built real-time inference pipelines using Python and MediaPipe.
- Designed RESTful APIs using FastAPI for AI inference services.
- Integrated AI services into production web applications.
- Optimized inference latency for real-time deployment.

**Tech Stack:** Python | FastAPI | PostgreSQL | React | TypeScript | Tailwind CSS

## Details

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## Links

[LinkedIn](#)  
[Github](#)  
[Portfolio](#)

## Skill

- Backend: Python, FastAPI, Django, Node.js/Express, NestJS, REST API, gRPC, Microservices, WebSocket
- Frontend: React, TypeScript, Redux, Tailwind CSS, responsive UI
- Database: PostgreSQL, MySQL, MongoDB
- AI/ML & LLM: LangGraph/LangChain, RAG, LLM function calling, prompt engineering, PyTorch, OpenCV, computer vision, NLP
- Cloud & DevOps: Docker, CI/CD (GitHub Actions), Nginx, Linux, Git, GitHub/GitLab, Agile
- AI Development Tools: Claude, ChatGPT, Grok

## Languages

English

## Hobbies

Sport, Arts

## Driver Behavior Detection System – Joint Project with MIT Research Lab, Enclave

JAN 2025 – PRESENT

- Designed scalable data preprocessing pipelines for large-scale driving datasets.
- Built feature extraction workflows for AI model training.
- Trained and optimized deep learning models for driver behavior detection.
- Evaluated model performance and improved prediction accuracy.
- Collaborated with research engineers to deliver production-ready AI solutions.

**Tech Stack:** Python | Pytorch | OpenCV | PostgreSQL

## Chatbot Module for Financial Management System – Alameda County Probation Department, Enclave

JUN 2024 – AUG 2025

- Developed an enterprise chatbot using Rasa as the conversational orchestration framework.
- Integrated Retrieval-Augmented Generation (RAG) and LLMs through custom Rasa actions to generate context-aware responses from enterprise knowledge.
- Built custom action servers to retrieve business data, invoke backend APIs, and orchestrate AI-powered workflows.
- Designed NLP pipelines for intent classification and entity recognition using Rasa.
- Developed backend REST APIs using Django and deployed production AI services with Apache2.

**Tech Stack:** Python | Django | Rasa | LLM | RAG | MySQL | Apache2

## 360° Performance Review System, Enclave

MAR 2023 – MAY 2024

- Built 360° Performance Review module in a sub-team of 4 (within 8-member team)
- Developed backend microservices using NestJS and gRPC for inter-service communication, real-time reviewer notifications using WebSocket
- Developed frontend using React, Tailwind CSS, React-Redux
- Enabled multi-language support (i18n) to support international users
- Containerized services using Docker to ensure consistent development and deployment environments.

**Tech Stack:** TypeScript | React | Tailwind CSS | Redux | NestJS | gRPC | MongoDB | WebSocket | Docker

## E-commerce Website for a Japanese Supermarket, Freelance

AUG 2024 – AUG 2025

- Developed full-stack e-commerce applications using React, Laravel, and MySQL.
- Built customer and admin portals with product management, shopping cart, online payment, and order management.
- Designed and integrated REST APIs between frontend and backend.
- Implemented responsive UI based on Figma designs.
- Optimized SQL queries and database structure using MySQL.
- Deployed and maintained production applications on Hostinger.

**Tech Stack:** PHP | Laravel | React | TypeScript | Redux | TailwindCSS | MySQL

## NGO Management System, Enclave

MAR 2022 – MAR 2023

- Maintained and enhanced enterprise PHP web applications.
- Developed new backend features and REST APIs.
- Designed and optimized SQL queries.
- Fixed legacy issues and improved application performance.
- Collaborated with cross-functional teams to deliver new features.

Tech Stack: PHP | JavaScript | HTML/CSS | SQL Server

## Education

Bachelor of Engineering in Information Technology, Da Nang University of Science and Technology, Da Nang

2018 – 2024